Solar ESS Co. Ltd.
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ENERGY USE REPORT · JULY – OCTOBER 2026

Chicken farm, Chiang Rai (20 kWp)

Solar production, total consumption and night-time consumption · 20 kWp grid-tied system, no battery

PERIOD	DATA	ISSUED
1 July – 30 October 2026	Monitoring platform, 5-minute records	31 October 2026

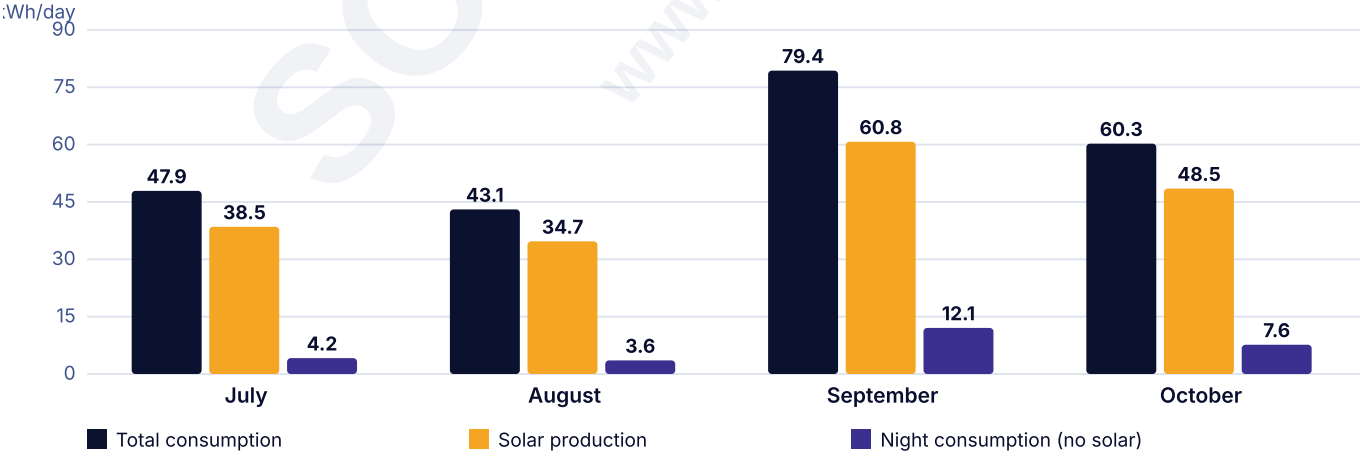
Solar production	Total consumption	Night consumption	Night share	Days with data
4,558	**5,758**	**694**	**12%**	**98**
kWh · 4 months	kWh · 4 months	kWh · all from grid	of all consumption	of 123 days

Monthly figures

Month	Days with data	Solar production	Total consumption	Night consumption	Night share	Solar / day	Use / day	Night / day
July	31	1,193.0	1,485.5	128.8	9%	38.5	47.9	**4.2**
August	13	451.1	559.9	46.4	8%	34.7	43.1	**3.6**
September	24	1,458.5	1,904.8	289.5	15%	60.8	79.4	**12.1**
October	30	1,455.5	1,807.6	229.0	13%	48.5	60.3	**7.6**
Total	**98**	**4,558.1**	**5,757.8**	**693.7**	**12%**	**46.5**	**58.8**	**7.1**

kWh · The data logger was offline from 14 August to 6 September. Monthly totals count only days with data; use the per-day columns to compare months.

Average per day, by month

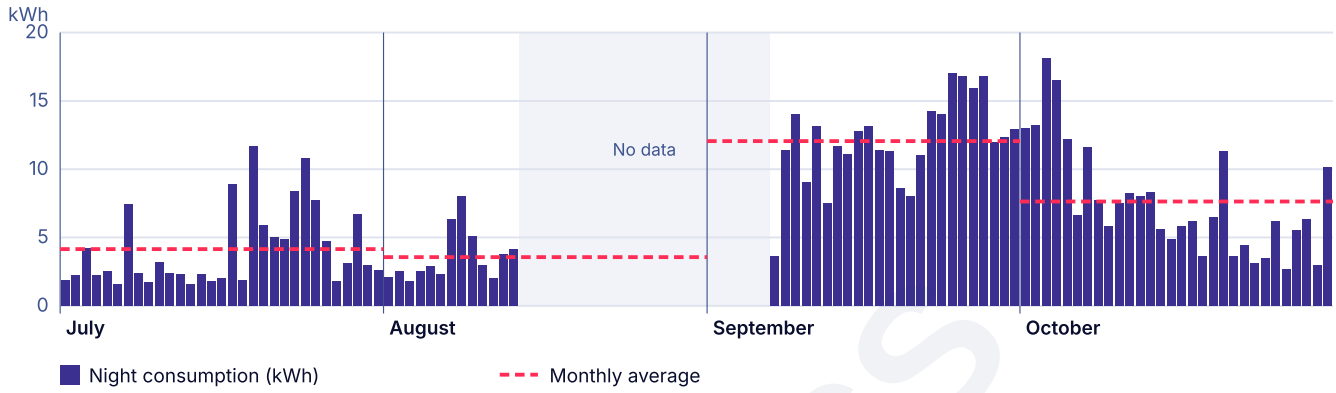


September was the busiest month: use jumped to about 79 kWh a day and night use tripled to 12 kWh. Solar production rose with the load, as the inverter's export limit keeps production in step with what the farm uses.

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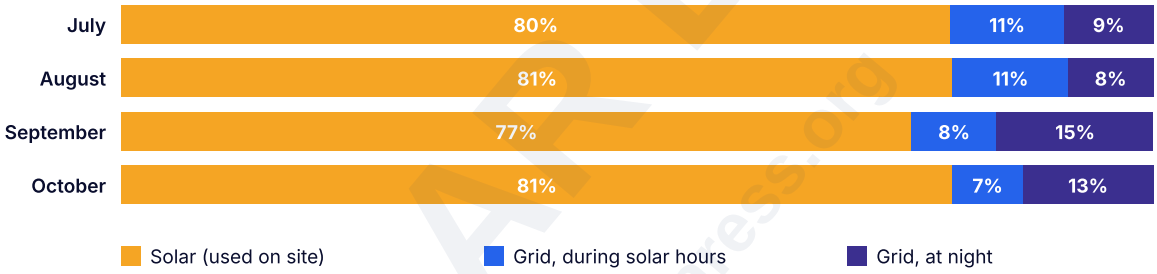
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Night-time consumption, day by day



Night = hours with no solar production that day. Night use was 2–5 kWh in July and August, 8–17 kWh in September and fell back to 3–13 kWh in October.

Where the farm's electricity came from



Share of each month's consumption. Every unit used at night is bought from the grid because the system has no battery.

Key findings

1. **Solar supplies most of the daytime need.** About 80% of the farm's electricity came from solar in July, August and October, and 77% in September.
2. **Night use is the gap solar cannot close.** 694 kWh were used at night over the four months, all bought from the grid: 4 kWh a night in July–August, 12 kWh in September, 7.6 kWh in October.
3. **September changed the pattern.** Day and night use both rose sharply, which points to more equipment running (for example heating, lamps, fans or pumps for a new flock).
4. **Monitoring gap.** No data was recorded from 14 August to 6 September. The logger connection should be checked so no period is lost again.

Recommendations

1. **Battery of about 10–15 kWh usable:** stores midday solar that the export limit now holds back and covers most nights (7.6 kWh at October level, 12 kWh at September level).
2. **Shift flexible loads to daytime** (09:00–15:00) where the farm can: pumps, water heating, feed equipment.
3. **Ask the farm what runs at night** on high nights (3–4.8 kW peaks) to confirm the load and right-size any battery.
4. **Check the data logger** Wi-Fi signal and power supply.

Notes: night and solar-hours consumption calculated from 5-minute data; totals agree with the platform's monthly figures within about 2%. October covers 1–30 October.